

Multi-Phase Results and Cross-Phase Analysis

Phase-Specific Findings

Foundation Phase (Phase 1) I

The configured foundation phase retained 46 records after its search, filtering, and de-duplication rules. Its taxonomy includes transit spectroscopy, emission spectroscopy, phase-curve analysis, and direct imaging. These labels describe the configured retrieval design; the synthetic fixture cannot establish how widely those approaches occur in the published literature.

JWST Phase (Phase 2) I

Phase 2 retained 17 records under the JWST-oriented queries and the configured year filter (≥ 2020). The boundary is intended to include pre-launch calibration work and post-launch observations. It is a design choice, not evidence that the phase is exhaustive or that any instrument caused a change in the field.

Molecular Detection Phase (Phase 3) I

Phase 3 retained 29 records under five configured molecule-oriented query groups: H₂O, CO, CH₄, H₂S, and Na/K. The query groups operationalize the review question; they do not assert that these are the most commonly detected species or that the fixture establishes a scientific trend.

Knowledge Graph Results

Knowledge Graph Results I

The optional knowledge-graph stage reports **0** assertions from the eligible sample. The configuration-derived hypothesis table is the authoritative mapping from review question to score:

Knowledge Graph Results I

ID	Hypothesis	Scope	Evidence score
H1	JWST Atmospheric Characterization	observational	pending
H2	Molecular Diversity Detection	chemical	pending
H3	Cross-Method Consistency	methodological	pending
H4	Theoretical-Observational Agreement	theoretical	pending

The configured scores are descriptive evidence summaries, not calibrated probabilities, causal effects, or conclusions about instrument performance. The table should be read together with the claim ledger and source-tier metadata. A fixture run cannot establish whether models and observations agree in the broader literature.

Citation Network Analysis

The combined corpus contains **132** citation relationships across **70** papers, with a network density of 2.73%. This is a descriptive statistic for the retained corpus; it does not establish a field-wide citation structure or community claim.

Reproducibility Assessment

Reproducibility assessment of the sampled papers revealed a mean composite reproducibility score of pending, with 0 papers successfully scored. This analysis examines the computational workflow transparency of each paper, evaluating source data availability, method documentation, experimental reproducibility, and output accessibility.